

A Uniform Trace-Envelope Criterion for Relative Regularized Determinants

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Abstract

We formulate the all-order theorem missing from the finite trace atlas. Let $\tau_n(\sigma)$ be the cloud-extracted complementary power traces. If

$$|\tau_n(\sigma)| \leq Mq^n, \quad n \geq 2,$$

uniformly in the noise, then for every R with $Rq < 1$ the logarithmic series converges uniformly and

$$\sup_{|z| \leq R} |\log R_\sigma(z)| \leq M[-\log(1 - Rq) - Rq] \leq \frac{M(Rq)^2}{2(1 - Rq)}.$$

The relative regularized determinants are consequently locally bounded and zero-free on that disk; Montel normality follows.

For the observed orders 2–12, the smallest unit-amplitude rates read directly from the maxima are $q_{\text{all}} = 0.35989$ and $q_{\text{fine}} = 0.14377$. The finite unit-disk log majorants are 0.07594 and 0.01074. If the same rates held at every later order, the all-order bounds would be 0.10117 and 0.01208. This last sentence is conditional: order thirteen and above are not controlled by the audit.

1 The trace formulation of the relative determinant

For a Hilbert–Schmidt complement, the regularized determinant has germ

$$R_\sigma(z) = \exp \left(- \sum_{n=2}^{\infty} \frac{\tau_n(\sigma)}{n} z^n \right).$$

The projection-free factorization of RH-234 identifies the same object from the complementary eigenvalue product [1, 2]. RH-235 shows why direct trace control can be much sharper than a Hilbert–Schmidt norm bound [3].

2 Uniform geometric envelope

Theorem 2.1 (Trace-envelope normality). *Let Σ be an index set and let $\tau_n(\sigma) \in \mathbb{C}$ for $\sigma \in \Sigma$ and $n \geq 2$. Suppose*

$$|\tau_n(\sigma)| \leq Mq^n$$

for constants $M \geq 0$ and $q \geq 0$, uniformly in σ . Then for every $R \geq 0$ with $Rq < 1$, the series

$$L_\sigma(z) = - \sum_{n=2}^{\infty} \frac{\tau_n(\sigma)}{n} z^n$$

converges uniformly on $|z| \leq R$ and satisfies

$$|L_\sigma(z)| \leq M[-\log(1 - Rq) - Rq] \leq \frac{M(Rq)^2}{2(1 - Rq)}.$$

Thus $R_\sigma(z) = e^{L_\sigma(z)}$ is a locally bounded zero-free normal family on $|z| < q^{-1}$.

Proof. For $|z| \leq R$,

$$\sum_{n=2}^{\infty} \frac{|\tau_n(\sigma)| |z|^n}{n} \leq M \sum_{n=2}^{\infty} \frac{(Rq)^n}{n} = M[-\log(1 - Rq) - Rq] \leq \frac{M}{2} \sum_{n=2}^{\infty} (Rq)^n.$$

The geometric sum gives the displayed bound and the Weierstrass test gives uniform convergence. Exponentiation gives $e^{-B} \leq |R_\sigma(z)| \leq e^B$, so the family and its reciprocals are locally bounded. Montel's theorem gives normality. $\square$

Corollary 2.2 (Locally uniform convergence criterion). *If, in addition, every fixed $\tau_n(\sigma)$ converges as $\sigma \rightarrow 0$, then R_σ converges locally uniformly on $|z| < q^{-1}$ to the determinant generated by the limiting traces.*

Proof. Split the logarithmic series into a finite head and a uniformly small geometric tail. The finite head converges coefficientwise; then exponentiate. $\square$

Theorem 2.3 (Quantitative coefficient-to-determinant convergence). *Under the hypotheses of Theorem 2.1, suppose $\tau_n(\sigma) \rightarrow \tau_n^{(0)}$ for each fixed n . For $Rq < 1$ and every $N \geq 2$,*

$$\begin{aligned} \sup_{|z| \leq R} |L_\sigma(z) - L_0(z)| &\leq \sum_{n=2}^N \frac{R^n}{n} |\tau_n(\sigma) - \tau_n^{(0)}| \\ &\quad + \frac{2M(Rq)^{N+1}}{(N+1)(1-Rq)}. \end{aligned}$$

In particular the first line controls a finite coefficient head and the second line is a uniform, explicitly summable tail.

Proof. The limiting coefficients inherit $|\tau_n^{(0)}| \leq Mq^n$. Split the difference at order N and bound each tail coefficient difference by $2Mq^n$. For $n \geq N+1$, use $1/n \leq 1/(N+1)$ and sum the geometric series. $\square$

If $B_R = M[-\log(1 - Rq) - Rq]$, the elementary exponential estimate also gives

$$\sup_{|z| \leq R} |R_\sigma(z) - R_0(z)| \leq e^{B_R} \sup_{|z| \leq R} |L_\sigma(z) - L_0(z)|.$$

Thus the theorem is suitable for a future validated coefficient audit: a finite certified head plus an analytic tail bound produces a determinant error bar on a compact disk.

3 Observed finite envelope

For a finite set of orders, define the unit-amplitude observed rate

$$q_{[2,m]} = \max_{2 \leq n \leq m} \left(\sup_{\sigma} |\tau_n(\sigma)| \right)^{1/n}.$$

This is a descriptive statistic, not an all-order certificate.

Statistic	all endpoints	fine endpoints
Observed orders	2–12	2–12
$q_{[2,12]}$	0.359888	0.143772
Finite unit-disk majorant	0.075936	0.010737
Conditional all-order bound	0.101170	0.012071
First uncontrolled order	13	13

Table 1: Finite observations and conditional geometric continuations.

The finite majorants in the table sum the maximum at each order, even when those maxima occur at different endpoints. They are therefore slightly larger than the largest single-endpoint jets of RH-236, namely 0.075920 and 0.010662. This distinction makes the table a valid uniform finite-order majorant rather than a representative trajectory.

The global rate is set by the second moment. Higher observed orders remain subunit as well, but eleven points cannot exclude a later spike. In particular, fitting a line to the observed logarithms would not prove the hypothesis of Theorem 2.1.

4 What the criterion gives—and what it cannot identify

The theorem controls both R_σ and R_σ^{-1} . Every locally uniform subsequential limit on $|z| < q^{-1}$ is therefore holomorphic and zero-free there. This is exactly the analytic conclusion needed to prevent the unresolved complement from creating or destroying a compact-set divisor inside the controlled disk.

Normality is not identification. Multiplying every member by a fixed zero-free analytic factor whose logarithmic coefficients obey a compatible geometric bound changes the limiting coefficients while preserving the same type of criterion after a suitable adjustment of M . Likewise a selector that absorbs too many regular roots can produce an exceptionally small relative factor. The coefficient anchor must independently specify the limiting logarithmic germ before the determinant can be matched to the intended deterministic numerator.

The radius q^{-1} is also local. Nothing in the criterion continues the limit across that boundary or supplies the global growth order needed for an entire divisor comparison. Those are later Gate-A questions even after a unit-disk envelope is proved.

5 Dynamical target

The traces arise from noisy periodic-loop integrals at fixed positive noise. The plausible analytic route is to extract the moving peripheral cloud in the trace formula and prove a bound uniform simultaneously in orbit length and noise. Such a theorem must exploit dynamical cancellation; a generic trace-ideal estimate returns to the RH-229 wall.

Concretely, one may seek a grouped periodic decomposition

$$\tau_n(\sigma) = \sum_{\mathcal{G}} W_{\mathcal{G},n}(\sigma)$$

in which the cloud contribution is removed before estimating the grouped weights. An absolute bound on every ungrouped loop may be too expensive and recreate the Hilbert–Schmidt obstruction. The useful theorem is instead a cancellation-preserving majorant whose total is $O(q^n)$ uniformly as $\sigma \rightarrow 0$.

A proof may also be hybrid: certify the first N traces directly, derive a uniform long-orbit estimate for $n > N$, and use the quantitative theorem above to join the two regimes. This is more flexible than demanding a single sharp formula at every order.

Even a successful envelope does not identify the limit by itself. One also needs a coefficient anchor proving that the selected cloud removes the singular factor and leaves the intended deterministic numerator. These are the two next Gate-A walls. Gates B–E and every Riemann-hypothesis conclusion remain untouched.

References

- [1] Barry Simon. *Trace Ideals and Their Applications*. American Mathematical Society, second edition, 2005.
- [2] Liang Wang. Projection-free spectral factorization of the regularized determinant. 2026. RH-234 preprint.
- [3] Liang Wang. Trace powers versus hilbert–schmidt mass in nonnormal cloud complements. 2026. RH-235 preprint.